Pharmacology Basics

Introduction to Pharmacology

Learning Objectives

At the end of this chapter the student will be able to:

1. Define various terminologies used in Pharmacology
2. Understand pharmacodynamics and pharmacokinetics (absorption, distribution, metabolism and excretion) of drugs
3. Understand theoretical pharmacokinetics like half-life, order of kinetics, steady state and plasma concentration
4. Understand drug safety and effectiveness like factors affecting drug action and adverse drug reactions

Definitions

Pharmacology:

from Greek , pharmakon, "drug"; logia, "the study of"

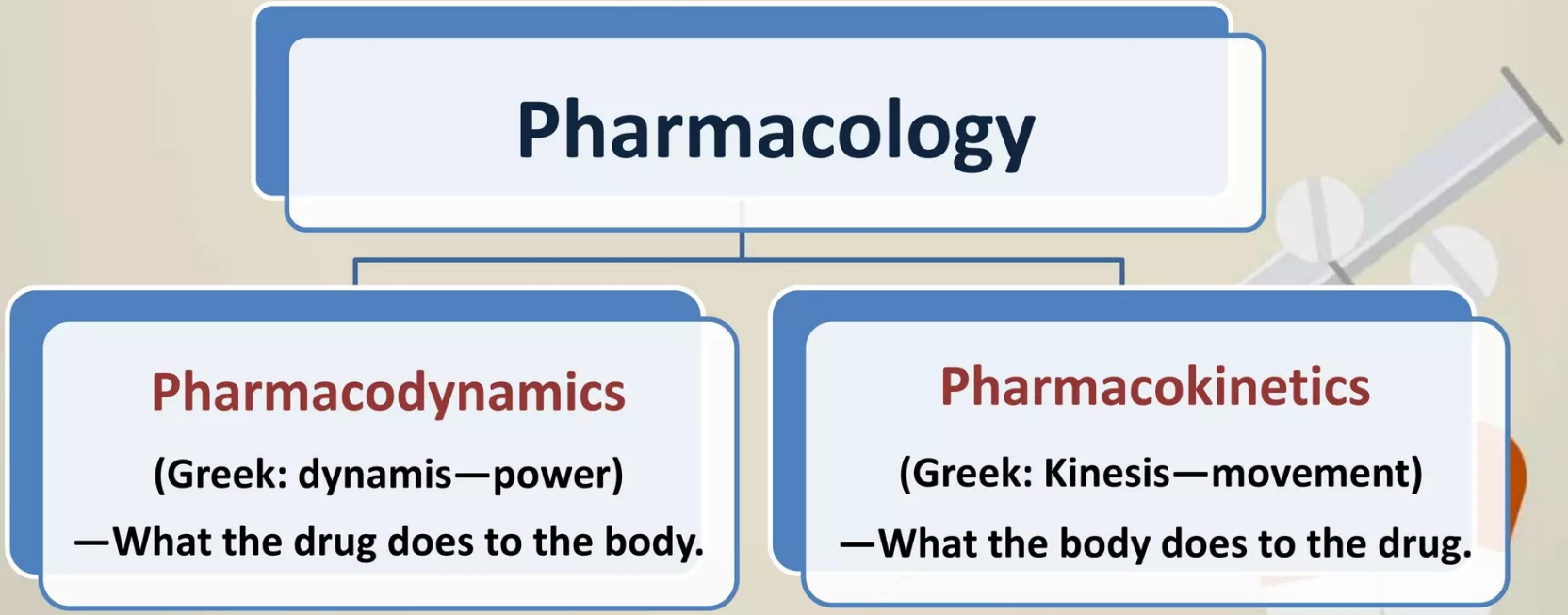
- The study of substances that interact with living systems through chemical processes
- It includes the history, source, physicochemical properties, dosage forms, methods of administration, absorption, distribution, mechanism of action, biotransformation, excretion, clinical uses and adverse effects of drugs

Drug

A substance which has a physiological effect when introduced into the body. Used in the prevention, treatment or diagnosis of a disease.

PHARMACOLOGY

- **Drug:** " A drug is any chemical molecule, which can be administered exogenously into the living system produce biological response or effect."
- ***Pharmacology*** is the science of drugs (Greek :*Pharmacon—drug; logos—discourse in).*



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Pharmacology is mainly divided into:

1. **Pharmacokinetics** (PK): Study of the absorption, distribution metabolism and excretion (ADME) of drugs (“ What the body does to a drug ”)
2. **Pharmacodynamics** (PD): The study of the biological and therapeutic effects of drugs (“What the drug does to the body”)
3. **Pharmacotherapeutics**: It deals with the proper selection and use of drugs for the prevention and treatment of disease
4. **Toxicology:** Toxicology is the branch of pharmacology dealing with the "undesirable" effects of drugs on biological processes

Pharmacology - All Branches

1. Pharmacokinetics
2. Pharmacodynamics
3. Therapeutics
4. Chemotherapy
5. Toxicology
6. Clinical Pharmacology
7. Pharmacy
8. Pharmacognesy
9. Pharmacogenetics
10. Pharmacoeconomics
11. Pharmacoepidemiology
12. Comparative Pharmacology
13. Animal Pharmacology
14. Posology

Food & Drug Administration (FDA)

- The federal agency responsible for the safety and efficacy of all drugs in the US. It is also responsible for food and cosmetics.

Adverse or “Side-Effect”

- An unintended action of a drug.
- Results from a lack of specificity of drug action.
- All drugs are capable of producing adverse effects.

Drug Names

Chemical name

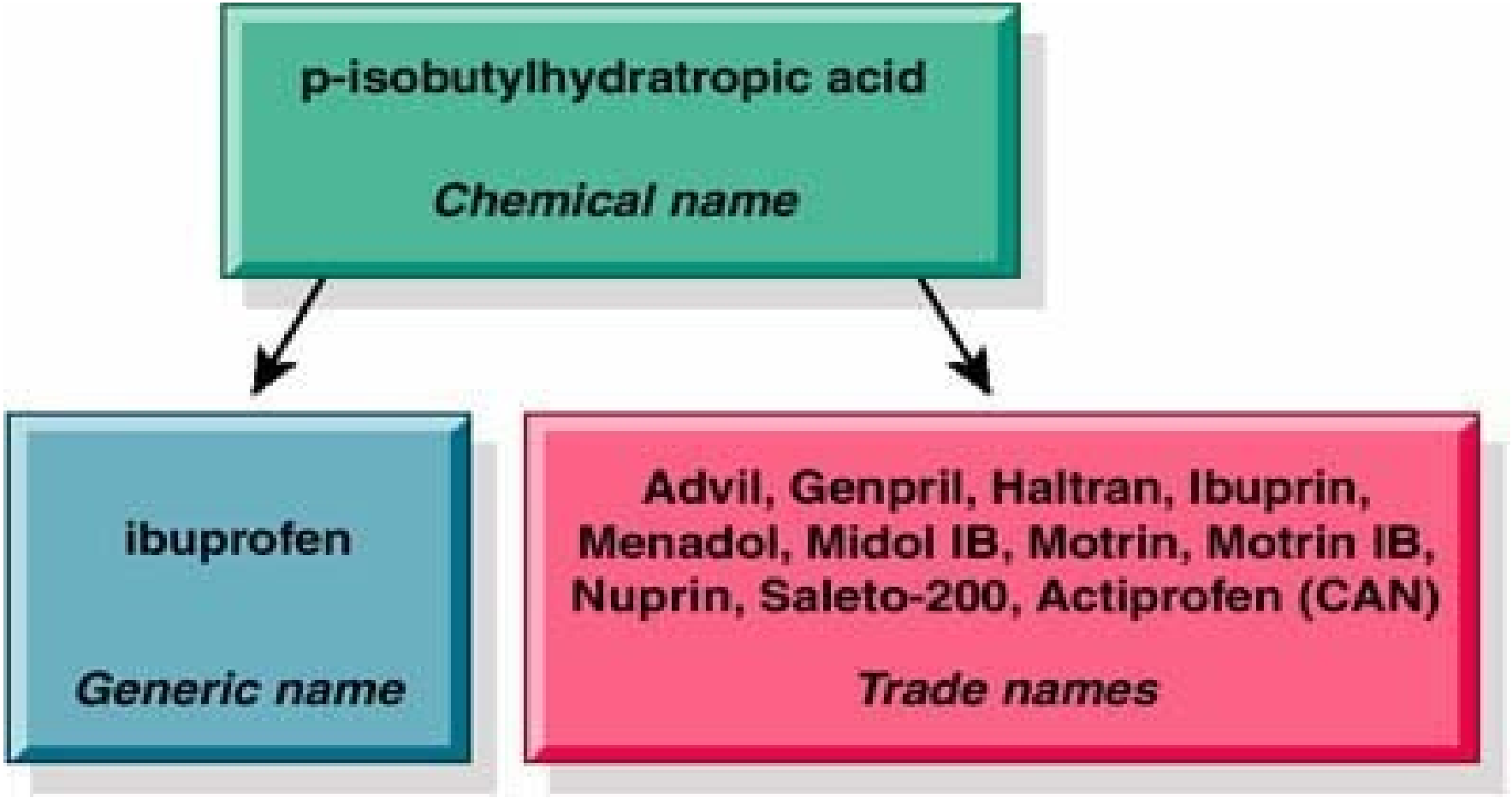
- The drug's chemical composition and molecular structure

Generic name (nonproprietary name)

- Name given by the United States Adopted Name Council

Trade name (proprietary name)

- The drug has a registered trademark; use of the name restricted by the drug's owner (usually the manufacturer)



- To avoid confusion: it's best to use a drug's generic name

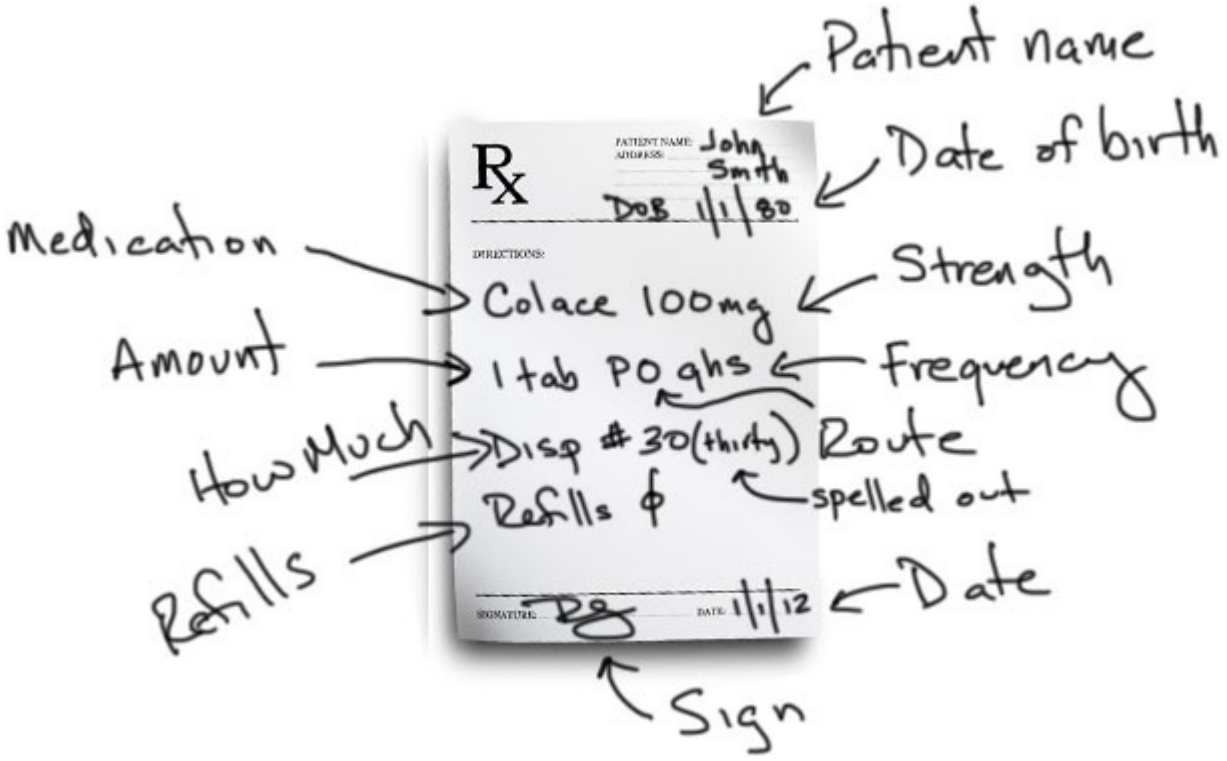
Grouping of drugs:

* Pharmacologic class (or family):
  - drugs that share similar characteristics
  - Example: beta-adrenergic blockers are an example of a pharmacologic class.

- Therapeutic class:
  - groups drugs by therapeutic use
  - Example: Antihypertensives

It's better to classify drugs based on their pharmacological properties

Principles of Prescription Writing



Definition

A prescription is a written, verbal, or electronic order from a practitioner or designated agent to a pharmacist for a particular medication for a specific patient.



The “Five Rights” of Medication Administration

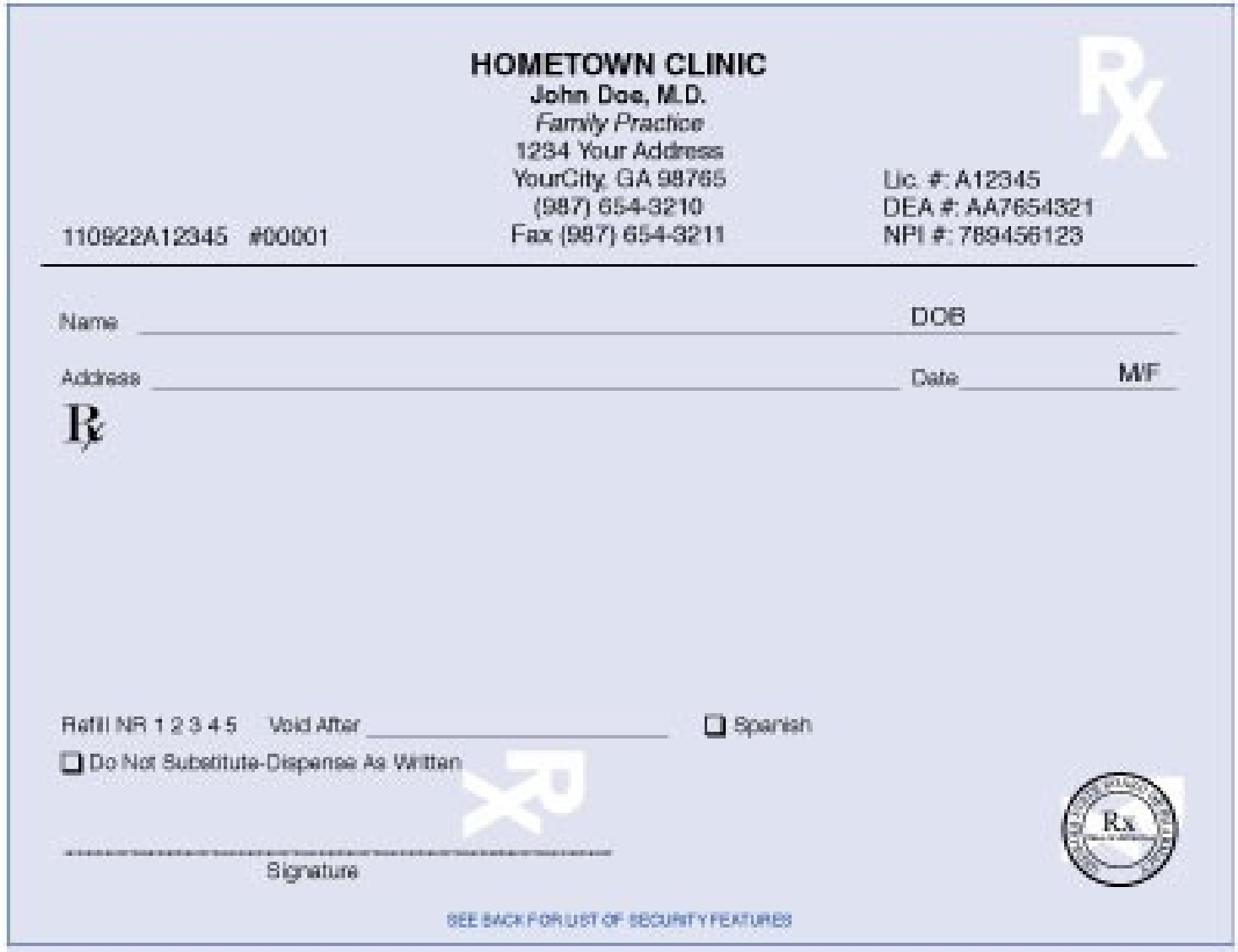
- Right drug
- Right dose
- Right time
- Right route
- Right patient

Prescription Formatting

Heading

Body

Closing



- Tablets - tab
- Capsule – cap
- Syrup – syr
- Suspension – susp
- Injection – Inj
- Metered dose inhaler – as such
- Lotion – as such

- PO: by mouth
- PR: per rectum
- IM: intramuscular
- IV: intravenous
- ID: intradermal
- IN: intranasal
- TP: topical
- SL: sublingual
- BUCC: buccal
- IP: intraperitoneal

Recent recommendations : don't use the following abbreviations

Latin abbreviations frequently used in prescription writing

Abbreviation	*Latin*	*English*
ad lib.	*ad libitum*	at pleasure
a.c.	*ante cibum*	before meals
aq.	*aqua*	water
b.i.d.	*bis in die*	twice a day
caps.	*capsula*	capsule
c̄	*cum*	with
d.	*dies*	a day, daily
disp.	*dispensa*	dispense
gtt.	*guttae*	drops
h.	*hora*	hour
h.s.	*hora somni*	at bedtime
non rep.	*non repetatur*	do not repeat (or refill)
no.	*numerus*	number, amount
p.c.	*post cibum*	after meals
p.r.n.	*pro re nata*	as needed
q.h.	*quaque hora*	every hour
q. 4 h.	*quaque quarta hora*	every 4 hours
q.i.d.	*quater in die*	four times a day
Sig.	*signa*	let it be labeled, label
stat.	*statim*	immediately
tab.	*tabella*	tablet
t.i.d.	*ter in die*	three times a day

Classification of Drugs

based on prescriptions

- General Sales List.
- Pharmacy Medicines, Over The Counter (OTC).
- Prescription Only Medicines (POM).
- Controlled Drugs (CD).

Controlled Substances

- Definition - a prescription drug whose use and distribution is tightly controlled because of its abuse potential or risk
- Regulation is more strict

Routes of Drug Administration

The route of administration (ROA) that is chosen may have a profound effect upon the speed and efficiency with which the drug acts

ROUTES OF DRUG ADMINISTRATION

How the drug is given

Enteral

1. Oral
2. Sublingual
3. Rectal



Parenteral (injectable)

1. Intravenous
2. Intramuscular
3. Subcutaneous



Topical

1. Intranasal
2. Inhalation
3. Intravaginal



ROUTES OF DRUG ADMINISTRATION

- The route of administration is determined by:
  1. Properties of the drug (for example, water or lipid solubility, ionization)

  2. Therapeutic objectives (for example, the desirability of a rapid onset of action, the need for long-term treatment, or restriction of delivery to a local site).
- Major routes of drug administration include:
  A. Enteral
  B. Parenteral
  C. Others: transdermal, nasal inhalation, oral inhalation, topical (skin cream, eye drops, ear drops), vaginal … etc

I. Enteral

- Enteral administration is
  - Drug placed directly in the GI tract:

- When the drug is given in the mouth, it may be:
  - Swallowed → allowing oral delivery (oral)
  - It may be placed under the tongue (sublingual) or between the gums and cheek (buccal)

    → direct absorption into the bloodstream
  - Rectal: Absorption through the rectum

1. Oral:

- **Advantages:**
  - **Convenient -** can be self- administered, pain free, easy to take
  - **Absorption** - takes place along the whole length of the GI tract
  - **Cheap** - compared to most other parenteral routes
  - **Safe-** low risk of infection
  - Overdose and toxicities can be overcome by antidotes, such as charcoal

1. Oral:

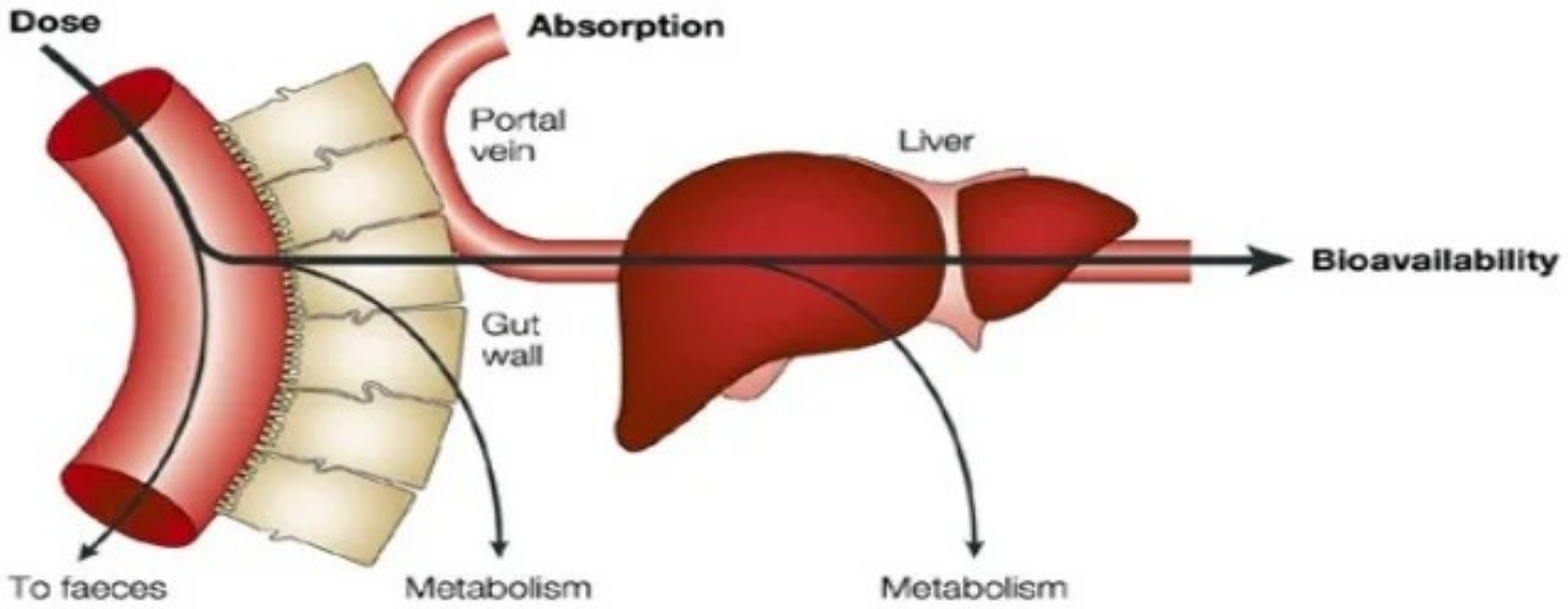
- Disadvantages:
  - Sometimes inefficient - only part of the drug may be absorbed
    - Absorption can be affected by several factors, as gastric pH
    - First-pass effect - drugs absorbed orally are initially transported to the liver via the portal vein
    - Destruction of drugs by gastric acid and digestive juices
  - Irritation to gastric mucosa - nausea and vomiting
  - Effect too slow for emergencies
  - Unpleasant taste of some drugs
  - Unable to use in unconscious patient

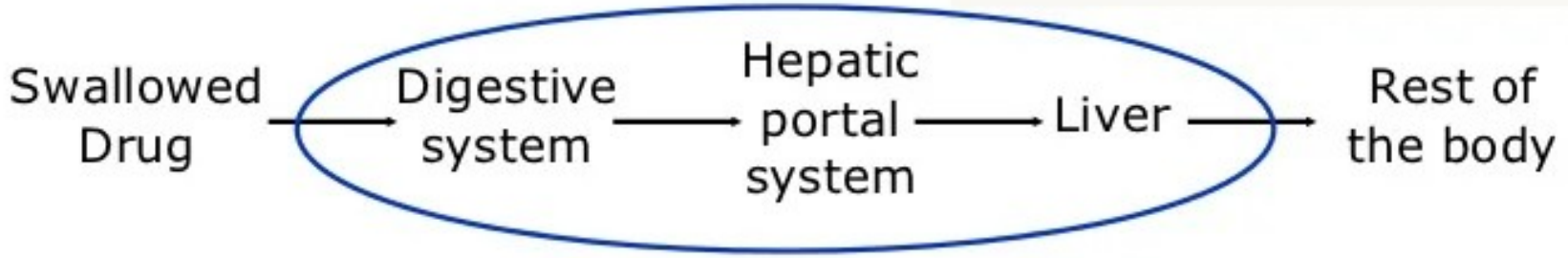
First-pass Effect

- The first-pass effect is the term used the intestinal and hepatic degradation or alteration of a drug or substance taken by mouth, after absorption, removing some of the active substance from the blood before it enters the general circulation

- The greater the first-pass effect, the less the agent will reach the systemic circulation when the agent is administered orally

First Pass Metabolism





1. Oral:

- Wide range of oral preparations including:
  A. enteric-coated
  B. extended- release preparations

A. Enteric-coated preparations:

Is a chemical envelope that resists the action of fluids and enzymes in the stomach but dissolves readily in the upper intestine.

- Such coat is useful for:
  1. Drugs that are acid unstable (e.g. Omeprazole)
  2. Stomach irritating drugs (e.g. Aspirin)



B. Extended-release Preparations (ER or XR):

- Have special coating or ingredients that control how fast the drug is released from the pill into the body

Advantages:

- Slower adsorption and prolonged duration
- Better patient compliance
- May maintain concentrations within acceptable therapeutic range over a long period of time
- These preparations are advantageous for drugs with short half lives

Ex, Morphine:

- **Its half-life is 2 -4 hours in adults**
- **oral morphine has to be administered 6 times in24 hrs to obtain continuous analgesic effect.**
- **Controlled-release tablets allow twice daily dosage**

2. Sublingual/ Buccal:

Placement under the tongue allows a drug to diffuse into the capillary network and enter the systemic circulation directly

- The buccal route: drug is place between cheek and gum

Advantages:

- Ease of administration
- Rapid absorption
- Low risk of infection
- Bypass of the harsh GI environment
- Avoid the first-pass metabolism



2. Sublingual/ Buccal:

- **Disadvantages**
  - Inconvenient
  - Small doses
  - Unpleasant taste of some drugs



3. Rectal

Advantages

- Unconscious patients and children
- If patient is nauseous or vomiting
- Good for drugs affecting the bowel such as laxatives
- Easy to terminate exposure

Disadvantages:

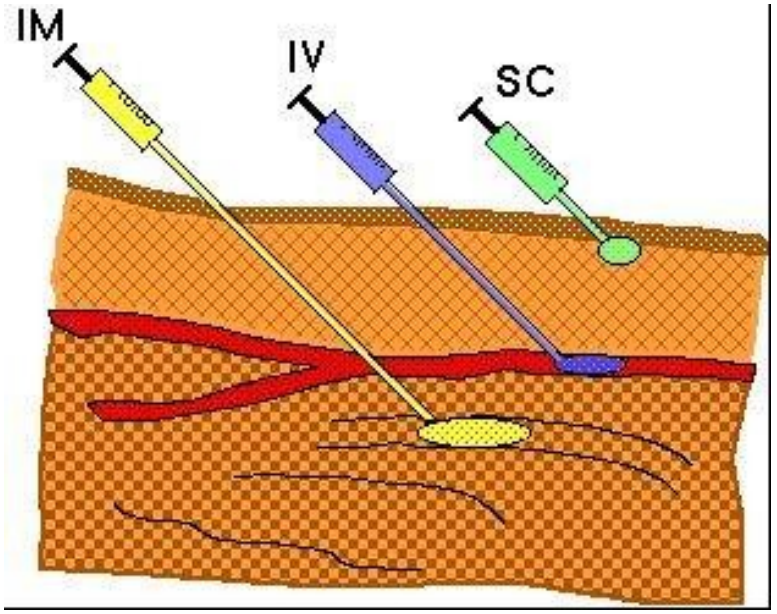
- Absorption may be variable
- Irritating drugs contraindicated

II. Parenteral route:

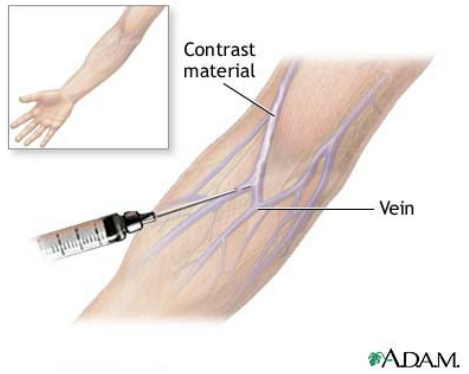
Introduces drugs directly across the body's barrier defenses into systemic circulation

- Uses:
  - Drugs that are poorly absorbed from GI tract (e.g. heparin).
  - Drugs that are unstable in the GI tract (e.g. insulin)
  - In patients unable to take oral medication (e.g. unconscious patients)
  - Circumstances that require a rapid onset of action

- Three major parenteral routes are:
  - Intravascular (IV, IA)- placing a drug directly into the blood stream
  - Intramuscular (IM) - drug injected into skeletal muscle
  - Subcutaneous - Absorption of drugs from the subcutaneous tissues



1. Intravenous (IV):



Uses and advantages:

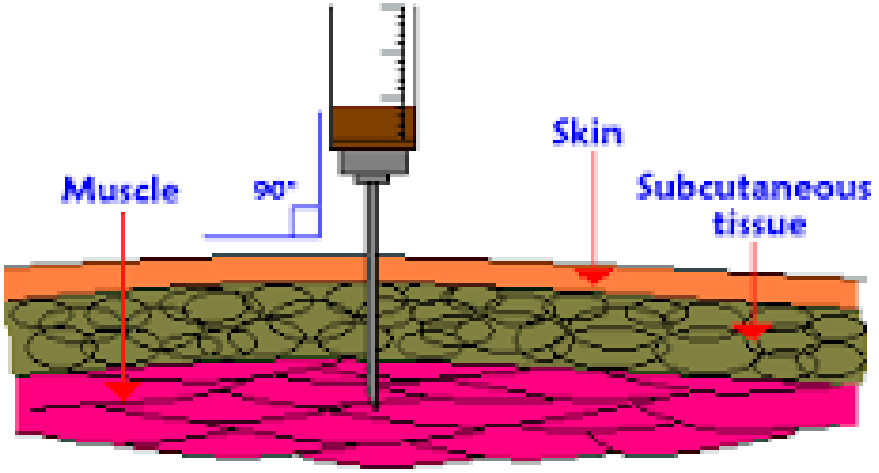
- For drugs that are not absorbed orally
- For administering chemicals that may cause irritation when administered via other routes
- Control over the amount of drug delivered
- Absorption phase is bypassed (100% bioavailability)
- Precise, accurate and almost immediate onset of action
- large quantities can be given, fairly pain free

1. Intravenous (IV)- cont

Disadvantages:

- Cannot be recalled by strategies, such as by binding to activated charcoal
- May introduce infection at the site of injection
- Risk of embolism
- Risk of adverse reactions by the too-rapid delivery of high concentrations of a drug to the plasma and tissues

2. Intramuscular (IM):



- Drugs administered IM can be:
  1. Very rapid absorption of drugs in **aqueous solution**

  2. Slow release preparations: **Depot Effect**

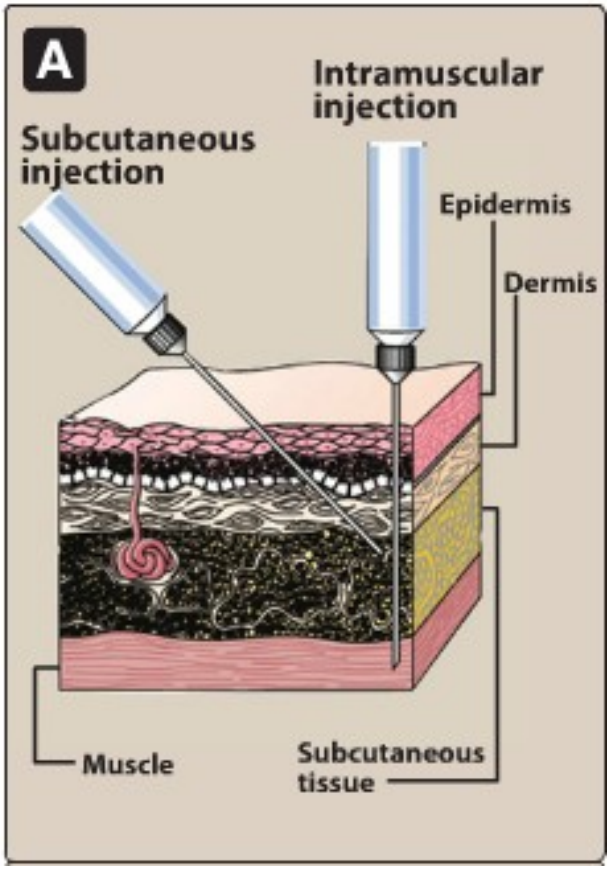
- Pain at injection sites for certain drugs

Depot preparations:

- Often consist of a suspension of the drug in a non-aqueous vehicle such as polyethylene glycol
- Vehicle diffuses out of the muscle, the drug precipitates at the site of injection
- The drug then dissolves slowly, providing a sustained dose over an extended period of time

3. Subcutaneous (SC):

- This route provides absorption via simple diffusion
- Slower than the IV route
- Advantages of SC injection:
  - Minimizes the risks of hemolysis or thrombosis associated with IV injection
  - May provide constant, slow, and sustained effects.
- SC route should not be used with drugs that cause tissue irritation
- Absorption is limited by blood flow, affected if circulatory problems exist



III. Others

1. Oral inhalation:

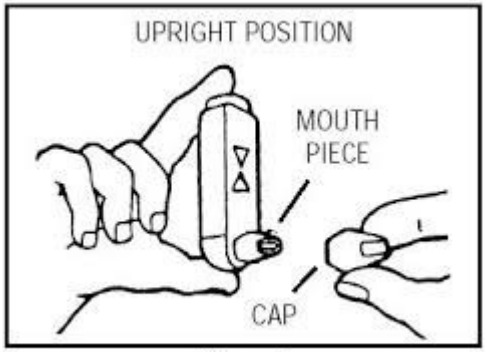


Figure 1



Figure 2

- Gaseous and volatile agents and aerosols
- Rapid onset of action due to rapid access to circulation
    a. Large surface area
    b. Thin membranes separates alveoli from circulation
    c. High blood flow
- This route is particularly effective and convenient for patients with respiratory complains (such as asthma)

2. Nasal inhalation:

- Administration of drugs directly into the nose.
- Agents include:
  - Nasal decongestants, such as oxymetazoline
  - Anti inflammatory corticosteroids, such as mometasone furoate

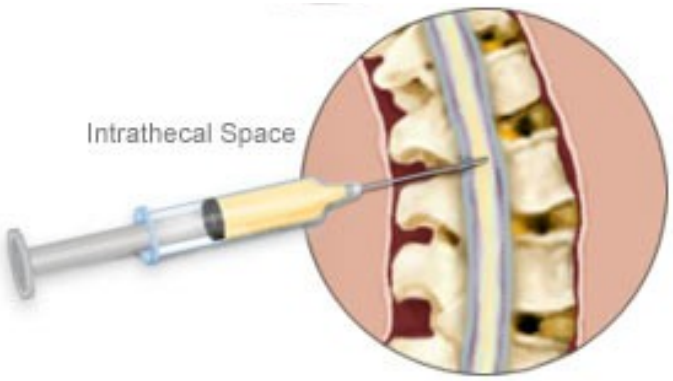


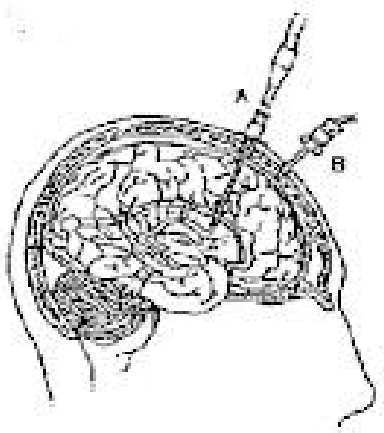


3. Intrathecal/intraventricular:

Introduce drugs directly into cerebrospinal fluid, when:

- Drugs cannot cross the blood-brain barrier to the CNS
- Local, rapid effects are needed, it is necessary to introduce drugs directly into the cerebrospinal fluid.

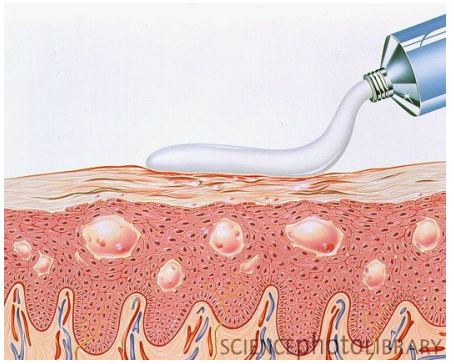




4. Topical:



- Topical application is used when a local effect of the drug is desired.
- Mucosal membranes (nasal, vaginal, etc.)
- Skin
  a. Dermal - rubbing in of oil or ointment (local action)
  b. Transdermal - absorption through skin (systemic action)
     I. Stable blood levels
     II. No first pass metabolism
     III. Drug must be potent or patch becomes to large
     IV. Variable rate of absorption



Important Info

The ROA is determined by the physical characteristics of the drug, the speed which the drug is absorbed and/ or released, as well as the need to bypass hepatic metabolism and achieve high conc. at particular sites